Witness: a sensor stream that carries its own proof and its own trust score

TSYP14 Ground Truth Under Attack, Track 2: False Data Injection. Phase 1 submission (anonymous).

1. Summary and track choice

Automated decisions in agriculture, insurance and water management trust environmental sensor streams. A single corrupted stream can silently trigger a wrong decision. We chose **Track 2 (False Data Injection)** because its question, "is this sensor stream consistent with its context?", is the most precisely testable: the standard test specifies the attacks (a 12% step bias, a gradual drift up to 15%, a seven-sample replay) on real series with hidden site and interval, so a detector can be measured honestly and defended under questioning.

Our system, **Witness**, combines two layers into one output. A **provenance layer** proves who produced each reading and that the sequence is complete. A **consistency layer** checks whether each reading is physically plausible given independent context, and produces a trust score that downstream systems can use to pass, de-weight or quarantine data.

Key idea: **a valid signature proves origin, not truth.** A compromised sensor signs its lies correctly, so cryptography alone is not enough. Conversely, consistency checks alone cannot prove a record was not replayed or altered in transit. Each layer covers the other's blind spot.

2. Threat model and relevance

Asset. A daily ground-sensor stream (we attack solar irradiance) feeding an automated decision such as a yield estimate, a heat or drought alert, or an index-insurance payout.

Attackers.

- **A1, outsider:** modifies or replays records in transit and has no key. Countered by the provenance layer.
- **A2, compromised sensor or gateway:** holds the signing key and signs any value. Countered by the consistency layer.

- **A3, adaptive attacker:** knows the detector and shapes the attack to stay under it (low-and-slow drift, noisy or season-matched replay). We do not claim to stop A3; we measure where detection breaks (section 6).

Harm. The attacker makes a consumer act on false data without anyone noticing. Harm grows with the time the corruption stays undetected, so we report detection delay as well as detection rate.

Assumptions and limits. Context sources are independent of the sensor and not compromised at the same time. Key management is out of scope. A very small, very slow bias is physically indistinguishable from natural variability.

Stated assumptions about the challenge (not yet confirmed by the organizers). The attacked variable is all-sky surface solar irradiance. Percentages are applied multiplicatively to the recorded value. Sentinel-2 is used as additional context through scene cloud cover.

3. Architecture and trust mechanism

Layer 1, provenance. Each reading is signed with HMAC-SHA256 over (sensor id, index, value, hash of the previous record). Records are hash-chained. The receiver rejects a bad MAC (tampering), a stale or repeated index (replay) and a broken chain (reordering or deletion). A software or emulated trust anchor is sufficient for this track. Measured overhead: about 7 microseconds to sign and 10 microseconds to verify per record, and 177 bytes per record.

Layer 2, consistency and trust score. Each day's reading is compared with what independent context predicts, evidence is accumulated over time, and a fusion model outputs P(compromised). We define trust = 1 - P/tau, where tau is the alarm threshold, and map it to PASS, DEWEIGHT (continuous) and QUARANTINE (with hysteresis so the state does not flicker).

Why both layers. In our layer-by-layer test, the crypto layer catches 100% of outsider attacks and 0% of compromised-sensor attacks, while the consistency layer is the only defence against the second case. On the hardest split of hidden sites, the consistency layer detects 36% of step biases, 21% of drifts and 50% of replays at about two false alarms per 1000 clean days.

4. Signal and data processing

- **Physics model.** A gradient-boosted model predicts the clear-sky index (irradiance divided by clear-sky irradiance) from temperature, humidity, rainfall and wind, expressed as anomalies against their own recent past, plus clear-sky irradiance. Working with the clear-sky index removes the seasonal and latitude geometry and makes the model transferable across sites.
- **Relative residual.** The reading divided by the prediction, minus one. Because the attacks are multiplicative, this is the natural quantity.
- **Context-dependent noise model.** Cloudy days are far noisier than clear days. A second model predicts the expected residual size from the context, so a 12% bias on a clear day (where the model is precise) counts as strong evidence, and on a very cloudy day as weak evidence. On real data, what mattered most was a context model that transfers across sites: using anomalies against recent history, instead of absolute levels that act as a site fingerprint, doubled cross-site model fit (R2 from about 0.2 to 0.44) and raised mean detection from 43% to 51% on development splits.
- **Temporal evidence.** CUSUM for step bias, an exponentially weighted average for slow drift, a 7-day rolling mean for short episodes.
- **Replay evidence.** Distance to the closest earlier 7-day window, plus a **twin test**: a natural look-alike window also has look-alike weather, a replay does not.
- **Sentinel-2 context.** Scene cloud cover from the last satellite pass and its age in days, aligned causally. Revisit gaps are handled by giving the model the age of the observation, so staleness is uncertainty, not an error. On real data it helped modestly: mean detection over the five site sets rose from 53% to 56% (step bias), 49% to 54% (drift) and 57% to 61% (replay), and the worst site set improved from 14% to 20%. The gain is small and comes from one run, so we treat it as supportive, not conclusive.

5. Detector design and evaluation

Fusion. A small monotone gradient-boosted classifier (more evidence never lowers the score) combines the evidence into P(compromised). The alarm threshold is set to a target false-alarm rate on held-out sites. The score is not a calibrated probability for a deployment prior; it is trained on a simulated attack prior.

Protocol. Sites are split with no overlap into five groups: physics model, noise model, fusion model, threshold setting, and hidden test sites. Attack positions are random. We compare detectors at a matched false-alarm rate and also report results at the deployed threshold.

Baseline. A univariate detector with the same machinery but with the expectation taken from the series' own recent history. It cannot see a bias that persists, because its reference moves with the data.

Results on the official reference data. The table below gives mean detection over five different sets of seven hidden sites (NASA POWER, 20 sites), at about two false alarms per 1000 clean days, with step biases of 6-20%, drift to 8-20% and a seven-day replay. On one split, the median detection delay was about 10 days for a step bias and about 40 days for drift. The detector runs in about 0.2 ms per day with under 0.5 MB of memory.

Detector	Step bias	Drift	7-day replay
Witness	53%	49%	57%
Univariate baseline	6%	9%	21%

Detection depends strongly on which sites are hidden: across the five site sets it ranges from 14% to 82% for Witness and from 4% to 35% for the baseline. A per-site analysis shows why. Detection is about 70-80% in clear-sky-dominated climates (Mediterranean, arid, dry-season monsoon), 20-30% where clouds dominate (maritime, high latitude), and about 4% in an equatorial convective climate. Weather context predicts irradiance well only where clouds are predictable, and that is the main limit of this approach.

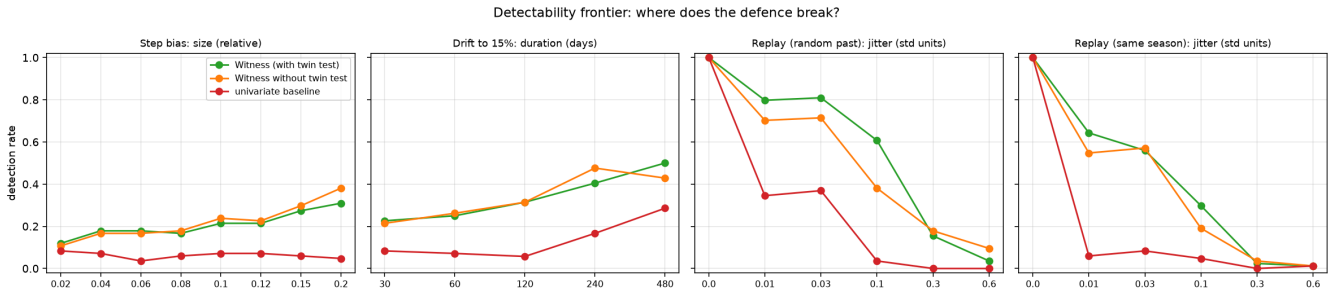
6. Standard scenario and evasion

Standard scenario. We inject a 12% step bias, a drift up to 15% and a seven-sample replay into real series at hidden sites and intervals. Results are in section 5: 53% step bias, 49% drift and 57% replay detection, against 6%, 9% and 21% for the baseline, at about two false alarms per 1000 clean days. Our random step sizes span 6-20%, so the figure is not the detection rate at exactly 12%.

Evasion analysis (the detectability frontier). Instead of a single accuracy number we sweep the attacker's choices and plot the detection rate:

- step bias size from 2% to 20%;
- drift duration from 30 to 480 days to reach 15%;
- replay with added noise, from a random past window and from the same season one year earlier.

The figure below sweeps the attacker's choices on the hardest split of seven hidden sites. An exact replay is always caught. With a little added noise, detection falls to about 80% (replay from a random past) and 55-65% (replay from the same season) at 0.01-0.03 standard deviations of noise, to 61% and 30% at 0.1, and to under 20% from 0.3 upward, so a noisy replay is the clearest evasion. Step-bias detection on this split is 12% at a 2% bias and 31% at 20%. Drift detection rises from 23% for 30-day drifts to 50% for 480-day drifts, because longer drifts leave more time to accumulate evidence. The twin test helps for noisy replays from a random past (61% against 38% at 0.1). The hardest case is replay with added noise. At about a third of the series' standard deviation, the copied values are statistically indistinguishable from natural weather variability, because the weather context explains under half of the variance on new sites; no detector built on this context can separate them. Better context, such as pixel-level Sentinel-2 data, is the route to closing this gap. The twin test improved this case but did not solve it. We report this as a limitation and as the first priority for Phase 2.



Held-out attack. We tested two attacks the detector never saw during training: a stuck sensor (frozen at its last good value for 10 days) and a zero-mean oscillating bias of 15% with a 14-day period. At the same false-alarm rate, the stuck sensor was detected in 32% of cases and the oscillating bias in 29%, against 6% and 10% for the univariate baseline. The detector generalises beyond its training attacks, but only partly: these two are harder than the standard attacks because they leave little persistent offset.

7. Feasibility

- Runs on a laptop with no purchased hardware or data: public NASA POWER data, public Sentinel-2 metadata, scikit-learn and numpy.
- The detector is causal and streaming: about 0.2 ms per day and under 0.5 MB. The replay search grows with history length; it can be bounded with a rolling window at a small cost in coverage.
- No radiated or live attack is used; all attacks are applied to recorded series.

- A deployment would place signing at the sensor or gateway and the detector at the consumer.

8. Reproducibility and clarity

- A single command reproduces every number and figure; the repository includes unit tests for the provenance layer, the attack simulator, the replay feature and the Sentinel-2 alignment.
- Fixed random seeds; site splits are documented.
- Anonymous repository: https://github.com/bakugouhaha75/Tsyp14Challenge

9. Limitations and Phase 2 plan

- Detection depends strongly on climate (about 4% in an equatorial convective climate against 70-80% in clear-sky climates). Same-season noisy replay and very slow small biases are only partly detected.
- The score is not a calibrated probability for a deployment prior.
- Key management and simultaneous corruption of sensor and context are out of scope.
- Phase 2 priorities: close the gap on noisy replay, extend held-out attack testing, and prepare the live demo with a signed stream, an injected attack and a trust-score timeline.